

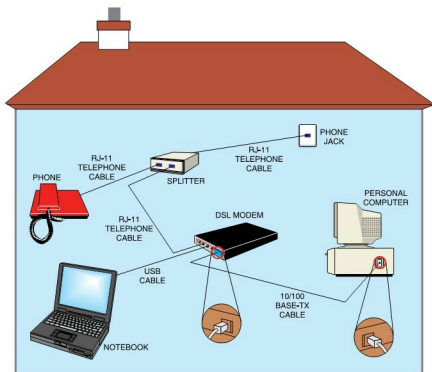


Fig. 2: A typical connection at subscriber's premises

ated by:

1. Speed of transmission
2. Maximum distance of signal transmission
3. Variation in speed between upstream and downstream
4. Symmetric or asymmetric character of the connection

Currently, the primary focus in xDSL is on the development and deployment of ADSL and VDSL technologies and architectures.

**ISDN digital subscriber line (IDSL).** This is an integrated services digital network (ISDN)-based technology that provides data flow rates of 144 kbps, which is slightly higher than the dual-channel ISDN data rate of 128 kbps. The goodness of IDSL lies in its 'always-on' connectivity which transmits data via a data network rather than the carrier's voice network. Thus it gives freedom from overloading of voice switches by data users.

IDSL uses a 2B1Q (two-binary, one-quaternary) line code on a single copper pair line to transmit information through the ISDN 'U' interface. The major limitation of IDSL is that customers cannot access ISDN signaling or voice

services. But if the requirement is Internet browsing at higher speed, IDSL is a better option than ISDN.

**High-bit rate digital subscriber line (HDSL).** HDSL is a symmetric technology that provides the same amount of bandwidth for upstream and downstream traffic. It offers speed of 2.048 Mbps over two copper pairs with operating distances of 3.6 km to 4.6 km, and is ideal for connecting PBX systems, digital local loops, point of presence, Internet servers and campus-based networks.

**HDSL-II.** HDSL-II, another technology proposed within the American National Standard Institute (ANSI) and the European Telecommunication Standard Institute (ETSI), offers the performance of HDSL but over a single pair.

HDSL was originally developed in USA, as a better technology for high-speed local exchange carrier systems which carry high-speed data links and voice channels using T1 lines. T-carrier circuits operate at 1.544Mbps and were carried using alternate mark inversion (AMI) line code. Due to limited range of AMI, the line code B8ZS (bipolar 8-zero

substitution) has been used. Later, 2B1Q line code was used, which allowed a 784kbps data rate over a single twisted-pair cable and 1.544Mbps with two twisted-pair cables. But, the problem still continued due to the differences between the T1 (1.544Mbps) and E1 (2.048 Mbps) standards.

A new standard for HDSL has been developed using the carrierless amplitude phase modulation (CAP) line code, which reached the maximum bandwidth of 2.048 Mbps using two pairs of copper. HDSL can be used either at the T1 rate or the E1 rate. Multiple of 64kbps channels inside the T1/E1 frame can be used to provide slower speeds to customers but the line rate is still the full T1/E1 rate.

HDSL further gave birth to two new technologies, called HDSL2 and SDSL. HDSL2 offers the same data rate over a single pair of copper and can work up to longer distances over a low-quality or lower-gauge copper. On the other hand, single-line rate digital subscriber line (SDSL) is a multi-rate technology offering speeds ranging from 192 kbps to 2.3 Mbps using a single pair of copper.

**Single-pair high-speed digital subscriber line (SHDSL).** Single-pair high-speed DSL technology supports symmetrical data rates. It is best suited for PBX, VPN, Web hosting and other data services that do not need the service guarantees of frame relay or the higher performance of a leased line. It cannot support voice service on the same pair as it takes over the entire bandwidth.

The ITU-T recommendation G.991.2 defines the standards for SHDSL. With one pair of copper line, the SHDSL having multiple of 64kbps payload provides symmetrical download and upload data rates ranging from 192 kbps to 2.304 Mbps. Moreover, the SHDSL provides symmetrical data rates from 384 kbps to 4.608 Mbps in 128kbps increments for two pair applications.

The distance covered is about 3 km and depends on the loop rate and noise conditions. One option to increase the coverage area is to decrease the data rates. Higher data rates can